

Name: _____ Date: _____

Period: _____

Planar Motion Check — KeyA particle moves with $f(t) = (t^2 - 1, 2t - t^2)$ for $0 \leq t \leq 3$.

1. Complete the table.

t	$x(t) = t^2 - 1$	$y(t) = 2t - t^2$	(x, y)
0	-1	0	(-1, 0)
1	0	1	(0, 1)
2	3	0	(3, 0)
3	8	-3	(8, -3)

2. Horizontal extrema:

 $x(t) = t^2 - 1$ is an upward parabola. Vertex at $t = 0$.**Minimum:** $x(0) = -1$ (leftmost position).**Maximum on $[0, 3]$:** $x(3) = 8$ (rightmost position).

3. When does the particle cross the
- y
- axis (
- $x = 0$
-)?

 $t^2 - 1 = 0 \Rightarrow (t - 1)(t + 1) = 0 \Rightarrow t = 1$ (only $t = 1$ is in $[0, 3]$).**At $t = 1$:** $(x(1), y(1)) = (0, 1)$. The particle crosses the y -axis at $(0, 1)$.**Teacher Note**

Q2: Students who report $t = 0$ as the minimum should be told to evaluate $x(0) = -1$ — the minimum *value* is -1 , not the time. Q3: $t = -1$ is not in the domain; accept only $t = 1$. Students who report $(0, 0)$ confused $y(0) = 0$ with the intercept coordinates — redirect them to evaluate $y(1) = 1$.